Foundational Verification of Probabilistic Programs

Mechanising the Lilac probabilistic separation logic in Lean

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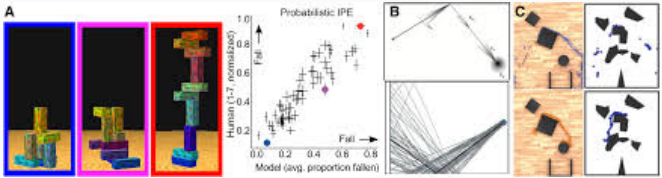
Probabilistic Programming is Everywhere

Probabilistic programming is everywhere

The most general feature of a PPL — sampling from a distribution — lives in the standard library of every major language.

- Cryptography — protocols, ZK proofs
- Differential privacy
- Randomised algorithms
- Distributed systems
- Bayesian inference & probabilistic ML
- Scientific simulation — climate, physics
- Robotics & perception — intuitive physics



Discrete vs. continuous

The dividing line that matters runs through the applications.

Discrete

Coin flips, Bernoulli / categorical choices, dice.

Cryptography, differential privacy, randomised algorithms — Applications traditionally within CS - much larger focus.

Continuous

Gaussian noise, sensor models, Bayesian priors over $\mathbb{R}$, physical quantities.

Scientific simulation, Bayesian inference, ML — where the real-world modelling happens.

Continuous distributions are where prior verification work fell short — and where this project makes its **first mechanised** contribution.

Verifying probabilistic programs is notoriously hard

- Conventional testing fails: a rare *correct* outlier execution and a genuine bug look *identical*.
- Correctness is a property of the whole distribution, not of any single run.
- Reasoning about independence and conditioning is subtle and error-prone by hand.

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So we need proof

Machine-checked, **foundational** arguments about distributions — resting on nothing but the axioms of mathematics.

Contribution

The first mechanisation of foundational verification

for probabilistic programs

supporting continuous distributions.

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1. Soundness proofs
2. Interactive *Proof Mode*

using Iris

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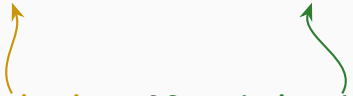
1. Soundness proofs
2. Interactive *Proof Mode* using Iris

Using a theorem prover (Lean/Rocq/Agda) Assuming only axioms of math[1]



The first mechanisation of foundational verification

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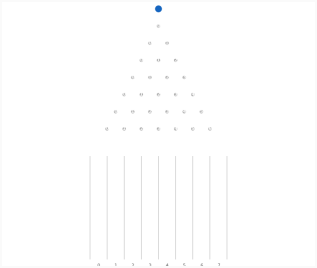
[1] Background Sec 2.4.1 “Dependent Type Theory”

A program is a distribution over runs

$$\text{flip} : (m : \mathbb{R}) \to \text{Distrib}\ \{m, m+1\}$$

an individual "run" of the program



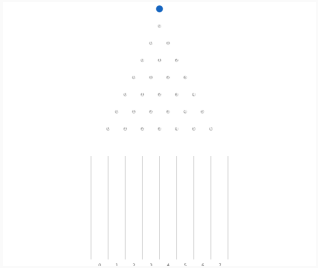
Honing in on a single marble

$$X_0 \leftarrow \text{flip } 0$$
$$X_1 \leftarrow \text{flip } X_0$$
$$X_2 \leftarrow \text{flip } X_1$$
$$\text{ret } X_2$$

$$|||$$

$$\text{for } (3,\ \text{flip } 0,\ \lambda X.\ \text{flip } X)$$

The program as a whole



All the marbles at once

Probabilistic semantics

$$
\begin{aligned}
&X_0 \leftarrow \mathsf{flip}\ 0\\
&X_1 \leftarrow \mathsf{flip}\ X_0\\
&X_2 \leftarrow \mathsf{flip}\ X_1\\
&\quad \mathsf{ret}\ X_2
\end{aligned}
\qquad \equiv \qquad M : \qquad [\![\Gamma]\!] \to \mathrm{Measure}\ ty
$$

Refining the domain: the Hilbert cube

$$
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&X_0 \leftarrow \text{flip } 0 \\
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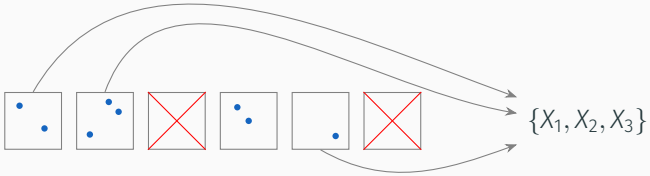
&\quad \text{ret } X_2
\end{aligned}
\quad \equiv \quad
M : \mathrm{HC} \to [\![\Gamma]\!] \to \mathrm{Measure}\ ty
$$

HC = Hilbert cube $= [0,1]^{\mathbb{N}}$



How to verify: probabilistic separation logic

$$\underbrace{X \leftarrow \mathrm{unif}[0,1];\ \ \mathrm{ret}\ X}_{\text{program}}$$

$$\equiv$$

$$\underbrace{\lambda E \mapsto \int (\lambda \mu \mapsto \mu\, E)\ d\big(\mathrm{map}\ (\lambda X \mapsto \mathrm{dirac}\, X)\ \mathrm{unif}[0,1]\big)}_{\text{denotation}}$$

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reason directly about the variables

$$X \sim \mathrm{unif}[0,1]$$

Iris, and the logic of bunched implications (BI)

The separation logic is built on the Lean port of Iris; its assertions Prop form a BI (bunched-implication) logic.

$\ulcorner \cdot \urcorner$	$: \mathrm{Prop} \to \mathrm{Prop}$	pure embedding
$\mathsf{True},\ \mathsf{False},\ \mathsf{emp}$	$: \mathrm{Prop}$	units
$\wedge,\ \vee,\ \Rightarrow,\ *,\ -\!\!*$	$: \mathrm{Prop} \times \mathrm{Prop} \to \mathrm{Prop}$	binary connectives
$\forall,\ \exists$	$: \forall A.\ (A \to \mathrm{Prop}) \to \mathrm{Prop}$	quantifiers

$*$ separating conjunction $-\!\!*$ magic wand — the separation-logic-specific connectives.

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Assertions Prop form a **BI (bunched-implication) logic** on the Lean port of Iris.

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- A logic inside a logic. Prop is an object logic inside Lean's meta-logic.[2]

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- A logic inside a logic. Prop is an object logic inside Lean's meta-logic.[2]
- Impredicativity. $\forall/\exists$ range over an *arbitrary* type A — including Prop itself: we may quantify over propositions.

[2] Revisited later in the talk.

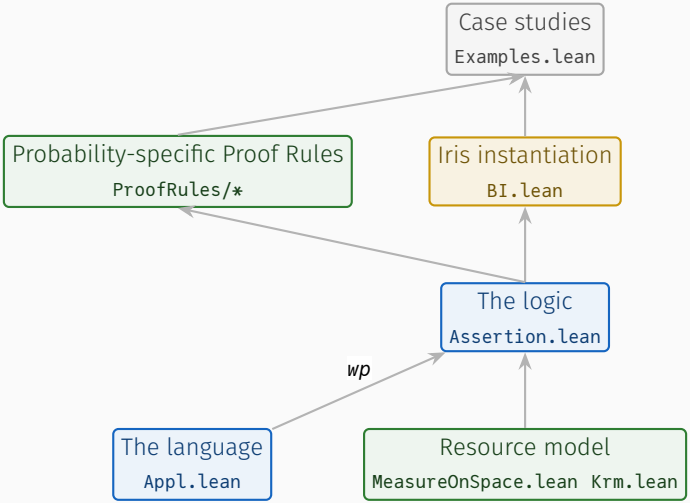
Mechanisation of Core[3] Lilac in Lean

a **continuous** probabilistic separation logic with separation as independence.

Along the way, plausibly the first (or second) instantiation of **Iris-lean** for a program logic of this kind.

[3] Extended Lilac adds reasoning about conditioning.

The formalisation, in one picture



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- Uniform: suppose (1) $\gamma, D, \mathcal{P} \vdash \forall_{\forall\alpha}\Sigma\mathcal{A}.\ X \rightsquigarrow \mathrm{Unif}[0,1] \rightarrow Q$. To show $\mathsf{wp}(\mathrm{Unif}\ [0,1], X; \mathcal{A}, Q)$, suppose $\mathcal{P}_{\mathrm{frame}} \bullet \mathcal{P} \sqsubseteq (\Sigma_\Omega, \mu)$ and $\mathrm{Dist} : \mathsf{RV}[\![\Lambda_{\mathrm{dist}}]\!]$. Let n witness $(D_{\mathrm{dist}}, \mathcal{P}_{\mathrm{frame}} \bullet \mathcal{P})$'s finite footprint. Write the Hilbert cube as $[0,1]^{\mathbb{N}} \cong [0,1]^n \otimes [0,1]^{\mathbb{N}}$. Define μ' via this isomorphism as the product measure $\mu|_{\Omega_n}{}^{\infty} \otimes \lambda$, where λ assigns to each finite-dimensional box $\prod_{i=1}^k [a_i, b_i] \times [0,1]^{\mathbb{N}}$ the measure $\prod_{i=1}^k |b_i - a_i|$ and extends to a measure on the whole Hilbert cube by the Carathéodory extension theorem. Let $\mathcal{P}_\omega$ be the restriction of μ' to measurable sets of the form $[0,1]^n \times F \times [0,1]^{\mathbb{N}}$. Let X be the projection $\pi_{n+1} = (\ldots, \omega_{n+1}, \ldots) \mapsto \omega_{n+1}$. By construction, the composite $\mathcal{P}_{\mathrm{frame}} \bullet \mathcal{P} \bullet \mathcal{P}_\omega$ is defined and $\mathcal{P}_{\mathrm{frame}} \bullet \mathcal{P} \bullet \mathcal{P}_\omega \sqsubseteq (\Sigma_\Omega, \mu')$ and X is $\mathcal{P}_\omega$-measurable and uniformly distributed in $[0,1]$. Therefore $\gamma, (D, X), \mathcal{P} \star \mathcal{P}_\omega \vDash \Theta$ by (1), and it only remains to show

$$\left(\begin{array}{l} \omega \leftarrow \mu; \\ v \leftarrow \mathrm{Unif}\ [0,1]; \\ \mathrm{ret}\ (D_{\mathrm{ext}}(\omega), D(\omega), v) \end{array}\right) = \left(\begin{array}{l} \omega \leftarrow \mu'; \\ \mathrm{ret}\ (D_{\mathrm{ext}}(\omega), D(\omega), X(\omega)) \end{array}\right)$$

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$$= \left(\begin{array}{l} (\omega_{1\cdots n}, v) \leftarrow \mu|_{\{1\cdots n\}} \otimes \mathrm{Unif}\ [0,1]; \\ \mathrm{ret}\ (D_{\mathrm{ext}}(\omega_{1\cdots n}), D(\omega_{1\cdots n}), v) \end{array}\right) = \left(\begin{array}{l} \omega_{1\cdots n+1} \leftarrow \mu'|_{\{1\cdots n+1\}}; \\ \mathrm{ret}\ (D_{\mathrm{ext}}(\omega_{1\cdots n}), D(\omega_{1\cdots n}), \omega_{n+1}) \end{array}\right)$$

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